

MyHomeUSB CRM — Deployment & Operations Guide

__VPS setup, env vars, deploy macro, troubleshooting, monitoring.__

1. Infrastructure

Layer	Component	Where
Host	Hostinger VPS	`187.127.152.103`
OS	Ubuntu 22.04	—
Webserver	Nginx	Listens on 80/443; reverse-proxies `/api` → 8001
Process manager	PM2	Manages `backend` (FastAPI / uvicorn)
Static	Nginx serves React build	`/var/www/myhomeusb/app/frontend/build`
DB	MongoDB	local on VPS, default port 27017
TLS	Let's Encrypt	auto-renew via certbot

Folder layout on VPS

```
/var/www/myhomeusb/app/  
  ■■■ backend/      FastAPI app  
  ■   ■■■ routes/  
  ■   ■■■ core/  
  ■   ■■■ server.py  
  ■■■ frontend/  
  ■   ■■■ src/  
  ■   ■■■ build/    Production bundle  
  ■■■ .env (root)   Shared env (not used at runtime)  
  ■■■ backend/.env  MONGO_URL, DB_NAME, JWT_SECRET, API keys  
  ■■■ frontend/.env REACT_APP_BACKEND_URL=https://myhomeusb.com
```

2. Environment Variables

`backend/.env` (required)

```
MONGO_URL=mongodb://localhost:27017  
DB_NAME=myhomeusb  
JWT_SECRET_KEY=<random>  
JWT_ALGORITHM=HS256  
JWT_EXPIRY_MINUTES=4320  
RESEND_API_KEY=<resend key, optional>  
GOOGLE_SHEETS_OAUTH_CREDS=<path to credentials.json, optional>
```

`frontend/.env`

```
REACT_APP_BACKEND_URL=https://myhomeusb.com
```

> **Never edit MONGO_URL/DB_NAME, REACT_APP_BACKEND_URL keys directly.** They're considered protected.
No defaults / fallbacks — missing values must fail fast.

3. Deploy Macro

From the agent's shell (after the user clicks **Save to GitHub** in the chat):

```
sshpass -p 'MyHome@VPS2026' ssh -o StrictHostKeyChecking=no root@187.127.152.103 \
```

```
"cd /var/www/myhomeusb/app && \
git pull && \
cd frontend && yarn build && \
pm2 restart backend"
```

What this does:

1. **git pull** — fetches the new commit from GitHub.
2. **yarn build** — produces **frontend/build/** static bundle (Nginx serves from here).
3. **pm2 restart backend** — bounces FastAPI to pick up any backend code changes.

After deploy:

- Verify the new commit hash matches expectations: `git log -n 1 --oneline`.
- **Hard-refresh** browser (Ctrl+Shift+R) to bust JS cache.

4. PM2 Commands

```
pm2 ls                # list managed processes
pm2 logs backend      # tail backend logs
pm2 restart backend   # graceful restart
pm2 stop backend      # stop
pm2 start backend     # start (uses ecosystem.config.js)
```

pm2-logrotate is enabled — logs rotate at 10 MB, retained for 30 days.

5. Nginx Configuration

/etc/nginx/sites-enabled/myhomeusb.com:

```
server {
    listen 443 ssl http2;
    server_name myhomeusb.com;
    ssl_certificate /etc/letsencrypt/live/myhomeusb.com/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/myhomeusb.com/privkey.pem;

    client_max_body_size 50M;

    # API → FastAPI
    location /api/ {
        proxy_pass http://127.0.0.1:8001;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_read_timeout 90s;
    }

    # SPA fallback
    location / {
        root /var/www/myhomeusb/app/frontend/build;
        try_files $uri $uri/ /index.html;
    }
}

server {
    listen 80;
    server_name myhomeusb.com;
    return 301 https://$host$request_uri;
}
```

Reload after edits:

```
sudo nginx -t && sudo systemctl reload nginx
```

6. MongoDB

Backup

`mongodump --uri="mongodb://localhost:27017" --db=myhomeusb --gzip --archive=/backups/$(date +%F).gz`

Recommended: cron job nightly + offsite copy to S3.

Restore

`mongorestore --uri="mongodb://localhost:27017" --gzip --archive=/backups/2026-02-15.gz`

Indexes (recommended)

```
db.users.createIndex({email:1}, {unique:true})
db.projects.createIndex({project_id:1}, {unique:true})
db.projects.createIndex({status:1, created_at:-1})
db.project_work_orders.createIndex({project_id:1, work_order_id:1}, {unique:true})
db.chèques.createIndex({status:1, cheque_date:-1})
db.cashbook.createIndex({project_id:1, payment_date:-1})
db.notifications.createIndex({user_id:1, is_read:1, created_at:-1})
db.material_requests.createIndex({project_id:1, status:1})
db.labour_expenses.createIndex({project_id:1, status:1})
```

7. Troubleshooting Playbook

Symptom	Likely cause	Fix
502 Bad Gateway	Backend not running	`pm2 restart backend`; check `pm2 logs backend` for tracebacks
504 Gateway Timeout	Slow Mongo query	Inspect query, add index, increase `proxy_read_timeout`
White screen on frontend	Stale JS bundle	Hard-refresh (Ctrl+Shift+R); rebuild + redeploy
"Insufficient permissions"	Role mismatch	Check `users.role`, log in/out, verify JWT contains role
Cheques showing "-" project	Pre-Feb-2026 row	API now backfills via project lookup — restart backend to pick up code change
RAB stuck "Pending PM"	PM hasn't acted	Check PM Approvals queue or notifications
"Cannot delete stage"	Stage has RAB	Reject/delete RAB first, then delete stage
"Unlock the section first"	Section locked	Toggle section unlock before touching item lock
Section visible on SE but not on Planning	Different `claim_type` filter	Planning's filter now matches `rework_se` + legacy `rework`; redeploy
Multi-stage bill releases as two payments	Backend missing `sibling_request_ids`	Verify latest backend (Feb 2026 `accountant_release_labour_payment`)
Cashbook reversal needed	Wrong release	Super Admin only; use audit log to find the entry

8. Monitoring & Alerts

Metric	Tool	Threshold
Backend uptime	PM2 + uptime-kuma	≤ 99.5% triggers alert
Mongo disk usage	df + cron	> 80% → alert
Suspicious logins	Audit log	> 5 failed in 5min → email Super Admin
Cheque bounce rate	Daily cron	> 2% → email GM
Backend error rate	logs grep	> 1% 5xx → page Super Admin

9. Public Documents (Architecture / Manuals)

After `yarn build`, these are served at:

- https://myhomeusb.com/MyHomeUSB_System_Architecture.pdf
- https://myhomeusb.com/MyHomeUSB_User_Role_Manual.pdf
- https://myhomeusb.com/MyHomeUSB_Business_Flow.pdf
- https://myhomeusb.com/MyHomeUSB_API_Reference.pdf
- https://myhomeusb.com/MyHomeUSB_Deployment_Guide.pdf

(Markdown counterparts also available at https://myhomeusb.com/MyHomeUSB_*.md.)

10. Routine Maintenance Schedule

Frequency	Task
Daily	Mongo backup + offsite sync
Daily	Cheque bounce rate check
Weekly	Log rotation review
Weekly	Index health (db.collection.getIndexes())
Monthly	Patch OS (sudo apt update && upgrade)
Monthly	Renew Let's Encrypt (auto, verify on 1st)
Quarterly	Restore-test from backup
Yearly	JWT secret rotation (force re-login)

End of ops guide.